

Bao Vo

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EDUCATION

University of Michigan, Ann Arbor, MI

Winter 2026

Master of Science in Electrical and Computer Engineering - Focus in Integrated Circuits & VLSI

- **GPA:** 3.68/4.00
- **Relevant Advanced Coursework:** Control Systems Analysis and Design, Integrated Analog/Digital Interface Circuits, Analog Integrated Circuits, Wireless/ML VLSI, VLSI Design II

University of Michigan, Ann Arbor, MI

Fall 2023

Bachelor of Science in Materials Science and Engineering (Major) and Electrical Engineering (Minor)

- **GPA:** 3.89/4.00 - Graduated Summa Cum Laude
- **Awards:** James W. Freeman Memorial Scholarship (2022-2023), James B. Angell Scholars (2023), Pan-Asia Alumni Scholarship (2019-2023), Clarence A. Siebert Memorial Scholarship (2023), Dean's List (2019-2023)
- **Relevant Advanced Coursework:** Monolithic Amplifier Circuits, VLSI Design, Analog Circuits, Digital Circuits

WORK EXPERIENCE

Project Manager and Veteran Life Science Consultant

Oct 2024 – May 2026

miLEAD Consulting | University of Michigan

- Developed market entry and pivot strategy for **3 pre-clinical biotechnology startups** (with highest valuation at **\$55M**) through primary research, conducting outreach to **2,000-4,000** potential consumers per project
- Utilized Pitchbook and market databases to present strategic recommendations directly to the leadership team on market positioning, pricing strategy, and target segments, driving pivots toward less saturated, higher-fit markets
- Collaborated with **U-M Coulter Partnership Program** to conduct market research, customer segmentation, and competitive landscape analysis for the university's early-stage innovations and proposals

Graduate Student Instructor

Aug 2025 – May 2026

Department of Electrical and Computer Engineering | University of Michigan

- Lectured, hosted office hours, and led laboratory sessions for EECS 215: Introduction to Electronic Circuits
- Developed laboratory materials on electronic circuits and amplifiers supporting instruction for **400+ students**

Graduate Student Researcher

Jul 2024 – Nov 2025

Wireless Integrated Circuits and Systems Group | University of Michigan

- Designed ultra-low-power circuits for implantable medical devices, focusing on a totally implantable auditory prosthesis featuring ultra-low-power subthreshold design with flip chip packaging
- Developed circuit blocks for the MEMS/ASIC interface, transimpedance amplifiers, and active RC filter for low-impedance, low-noise frontend system for an audio range application in TSMC 65nm
- Validated circuit stability and noise performance across varied PVT corners using Cadence Virtuoso and SPICE

DFT/STA Engineer

Jan 2024 – May 2024

Faraday Technology Corporation | Ho Chi Minh City, Vietnam

- Taped out an ultra-low-power chip in UMC 22nm for an ASIC shuttle project, driving chiptop DFT/STA flows to successful completion within an aggressive 1.5-month timeline while earning a top 10% performance evaluation
- Defined DFT specifications and strategy to meet customer test coverage targets and implemented the architecture from RTL to GDSII, including scan chain insertion, MBIST integration, simulation, and ATE patterns delivery
- Built and maintained automation flows (Tcl/Perl/Csh) for DFT and STA spanning pre-synthesis checks, ATPG, and post-layout verification, reducing manual effort and improving flow reliability
- Utilized PrimeTime to debug STA timing violations and provided recommendations to the physical design team
- Mentored senior undergraduates in a 2-month internship program, most of whom later converted to full-time roles

SKILL SET

- **EDA/CAD Tools:** Cadence Virtuoso, Layout, SPICE, Design Compiler, Innovus, PrimeTime, NCVerilog
- **Lab Tools:** Oscilloscope, Power Supply, Soldering Station, Multimeter, Prototyping
- **Software Skills:** Tcl, Perl, Csh, MATLAB, Python, C++, C, Verilog, SystemVerilog, Linux, Verdi, Git
- **Engineering Skills:** Hall Effect Measurement System, Dynacool Physical Properties Measurement System (PPMS), Scanning Electron Microscopy (SEM), Electron Dispersive X-Ray Spectroscopy, Tensile Tester
- **Nanofabrication Tool:** Angstrom Engineering Evovac Evaporator, Jetfirst150 RTP Rapid Thermal Process

PAST PROJECTS

Designing an NTT Accelerator for Post-Quantum Cryptography | University of Michigan

EECS 627

Jan 2026 – Apr 2026

- Designed a reconfigurable radix-2 NTT accelerator for polynomial sizes $n=256/1024$ using a Pease architecture with 16 parallel butterfly processing elements, resolving bandwidth bottlenecks via a 16-bank dual-ported SRAM array
- Implemented an AXI-4 subset-compliant DMA interface supporting out-of-order burst transactions (~3.6 GB/s total throughput) and integrated asynchronous FIFOs to manage cross-domain clock boundaries
- Completed synthesis and APR of the 10 mm × 8.4 mm full-chip design, securing timing closure at 100 MHz with a throughput of 298 kNTT/s

Designing a Quadrature Phase Mixer-First Radio Frequency Front-end | University of Michigan

Advisor: Professor David Wentzloff

May 2023 – Dec 2023

- Conducted literature research on modern differential ring oscillators and divide-by-2 frequency dividers for a low phase noise, low power receiver targeting IoT applications
- Modeled and conducted research on the novel Windmill divider for low-power quadrature-phase clock generation
- Designed a mixer-first quadrature phase front-end RF receiver using TSMC 65nm technology and Virtuoso ADE on Cadence, optimizing micro-level power consumption with good phase noise performance
- Optimized a second version of the receiver focusing on conversion gain and phase noise performance

Selected Analog and Digital Circuit Design Projects | University of Michigan

EECS 311, 312, 413, and 427

Jan 2023 – Dec 2023

- Project 1: Designed a high gain, low power three-stage bipolar transistor amplifier for bandpass filter study
- Project 2: Designed an 8-bit dual-mode ripple-carry adder in two optimizations for power and speed
- Project 3: Designed a low input/output impedance multi-stage amplifier with dual gain modes
- Project 4: Designed a low-voltage, low-power bandgap voltage reference
- Project 5: Designing a 16-bit reduced instruction set computing (RISC) processor with SRAM PUF

Designing and Power Characterizing Energy Harvesting Sensor Device | University of Michigan

Advisor: Professor David Wentzloff

Aug 2022 – May 2023

- Manufactured a circuit to perform electrical characterization of a commercial sensor click board
- Implemented C programs to monitor, record, and display the operation of the sensor on the integrated developing environment user interface
- Designed and manufactured a portable application board with a sensor based on its power profile. Integrated a power harvesting device into the board

Electrical Characterization of Novel Ultra-wide Bandgap Semiconductor | PetersonLab, SURE 2022 Project

Advisor: Professor Becky Peterson

May 2022 – Aug 2022

- Performed metal contact deposition on semiconductor thin films using physical vapor deposition with shadow mask technique, then conducted extensive experiments with the Lurie Nanofabrication Facility cleanroom and the Angstrom Evovac Engineering Evaporator
- Performed multi-temperature annealing and studied its effect on the film's conductivity and morphology
- Electrically characterized the semiconductor film using the Hall Effect Measurement System and Dynacool Physical Property Measurement System (PPMS)

HONORS & AWARDS

- **Ross +Tech Innovation Jam Competition - Track Winner and Finalist (2025):** Built ARTiculate, an AI-powered drawing therapy app for children with autism; led customer discovery, interviews, prototyping, and final pitch
- **Midwest Healthcare Case Competition Semifinalist (2025) – Team Leader** - CagriSema go-to market strategy
- **Southeast Case Competition Semifinalist (2026) – Team Leader** – Aurora Sparkling Juice expansion strategy
- **University of Michigan's Graduate Certificate in Innovation and Entrepreneurship (2026)**